

# Liliaokeawawa Cothren

Ph.D. Student, Electrical Engineering  
University of Colorado Boulder

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## Education

- 2021–Present **Ph.D. Electrical Engineering**, *University of Colorado Boulder*, Boulder, CO, Expected Graduation: Spring 2027.  
Advisors: Professor Elizabeth Bradley (CU Boulder) and Professor Raissa M. D’Souza (UC Davis)
- 2021–2023 **M.S. Electrical Engineering**, *University of Colorado Boulder*, Boulder, CO.
- 2017–2021 **B.S. Mathematics**, *Arizona State University*, Tempe, AZ.

## Honors and Awards

- 2026 Dynamics Days Conference 2026 Best Poster Award
- 2026 Dynamics Days Conference 2026 Travel Grant
- 2025 Selected as an “Editor’s Pick” for Chaos
- 2025 NSF ACCESS Allocations – Explore Level: Awarded 200,000 ACCESS Credits
- 2023 National Science Foundation Graduate Research Fellowship**
- 2022 Department of Electrical Engineering Travel Grant, University of Colorado Boulder
- 2021 Department of Electrical Engineering Travel Grant, University of Colorado Boulder
- 2021 Dean’s Future Leadership Fellowship (CU Boulder)
- 2021 ECEE Excellence Fellowship (CU Boulder)
- 2021 Graduate School Diversity Fellowship (CU Boulder)
- 2021 Graduated *Summa cum Laude*, Arizona State University
- 2018–2021 Dean’s List (ASU)
- 2020 Outstanding Junior for Wexler Students (ASU)
- 2017 New American University Scholar - President’s Award, Arizona State University

## Research Experience

- 2023–Present **Graduate Student Researcher**, *University of Colorado Boulder*, Advisor: Professor Elizabeth Bradley.  
Build core background in nonlinear systems theory to analyze and understand dynamics exhibiting catastrophic behavior. Verify theoretical findings with numerical simulations in Python. Apply theoretical concepts to relevant applications, including examples in ecology, economics, or social systems.
- 2021–2023 **Graduate Student Researcher**, *University of Colorado Boulder*, Advisor: Professor Emiliano Dall’Anese.  
Build core background in optimization and control theory to specifically tackle problems within data-driven control and concurrent learning via theoretical and algorithmic developments. Verify theoretical findings with numerical simulations in MATLAB or Python.
- 2019–2021 **Undergraduate Honors Thesis**, *Arizona State University*, Advisor: Professor Theodore Pavlic.  
Numerically simulate stochastic system via time synchronous clock and discrete-event-triggered codes. Formalize rigorous analysis proof that a simple decision-making heuristic guarantees optimal convergence onto a maximal caloric state of forager.

## Teaching Experience

- Fall 2026, **ECEN 3300: Linear Systems**, *University of Colorado Boulder*, Boulder, CO, Learning Assistant.
- Spring 2022 Topics include analysis of LTI systems in time and frequency domains and applications of linear systems, including communications, signal processing, and controls. Responsible for scripting and delivering regular review sessions, exam review sessions, and grading homework and exams.
- Fall 2023 **ECEN 5478: Online Convex Optimization**, *University of Colorado Boulder*, Boulder, CO, Learning assistant.  
Topics include basics of convex optimization and contraction theory. Responsible for hosting office hours, providing guidance on course projects, and grading homework and exams.

- Spring 2023 **ECEN 3300: Linear Systems**, *University of Colorado Boulder*, Boulder, CO, Learning Assistant. Topics include analysis of LTI systems in time and frequency domains and applications of linear systems, including communications, signal processing, and controls. Responsible for scripting and delivering regular review sessions, exam review sessions, and grading homework and exams.
- Spring 2020, Spring 2019 **IEE 380: Probability and Statistics for Engineers**, *Arizona State University*, Tempe, AZ, Teaching Assistant. Topics include discrete and continuous random variables and probability (mass or) density functions, hypothesis testing of means, variances, and proportions, and applications for engineering problems. Responsible for scripting and delivering regular homework review sessions, exam review sessions, and proctoring exams.
- Fall 2018 **FSE 100: Introduction to Engineering**, *Arizona State University*, Tempe, AZ, Teaching assistant. Topics include preliminary material to prepare students for engineering through a hands-on project focused on programming a robot to navigate a maze. Responsible for organizing materials and answering project questions.

## Professional Experience

- 2025—**The Math Lab at Front Range Community College**, Longmont, CO, Mathematics Tutor.  
Current Instruct and assist in answering questions for all math courses offered, including: Clinical Calculations, Math for Education, Liberal Arts Math, Statistics, College Algebra, Survey of Calculus, Trigonometry, Pre-calculus, Calculus I, Calculus II, Calculus III, Differential Equations and Linear Algebra.
- 2019-2021 **ASU Fulton Schools of Engineering Tutoring Centers**, Tempe, AZ, Lead tutor for mathematics and industrial engineering.  
Instruct and assist in answering questions related to industrial engineering, probability, statistics, calculus, linear algebra, ordinary differential equations, and real analysis. Write training curricula for tutors, including technical review sessions of frequent coursework and training on how to effectively tutor a variety of learning styles.
- 2019 **United Parcel Service (UPS)**, Phoenix, AZ, Industrial Engineering Intern.  
Collaborate on a multi-disciplinary team to plan for Peak season, including discussions with managers, plant engineers, and industrial engineers to draft accurate building layout and plot plans for 90 facilities. Presented recommendations in weekly updates, with success in redrafting some facilities' layouts for improved safety and efficiency.

## Research Interests

Nonlinear dynamics, catastrophe theory, control theory, early warning signals.

## Publications

### Journal papers

- [8] **L. Cothren**, E. Jensen, R. D'Souza, E. Bradley. "Controlling Coupled Cusp Catastrophes Using State-Dependent Ricatti Equations." Submitted to *IEEE Control System Letters*, 2026.
- [7] **L. Cothren**, F. Bullo, E. Dall'Anese. "Online feedback optimization and singular perturbation via contraction theory." *SIAM Journal on Control and Optimization*, 2026; Volume 64, Issue 3.
- [6] **L. Cothren**, R. D'Souza, E. Bradley. "Coupled catastrophes in systems with bidirectional feedback." *Chaos: An Interdisciplinary Journal of Nonlinear Science*, November 2025; Volume 35, Issue 11, Page 111105. **\*Selected as an "Editor's Pick."**
- [5] Y. Chen, **L. Cothren**, J. Cortes, E. Dall'Anese. "Online Regulation of Dynamical Systems to Solutions of Constrained Optimization Problems." *IEEE Control System Letters*, December 2023; Volume 7, pages 3789–3794.
- [4] **L. Cothren**, G. Bianchin, E. Dall'Anese. "Online Optimization of Dynamical Systems with Deep Learning Perception." *IEEE Open Journal of Control Systems*, September 2022; Volume 1, pages 306–321.

### Refereed Conference Proceedings

- [3] **L. Cothren**, G. Bianchin, E. Dall'Anese. "Data-Enabled Gradient Flow as Feedback Controller: Regulation of Linear Dynamical Systems to Minimizers of Unknown Functions." *4<sup>th</sup> Annual Learning for Dynamics & Control Conference*, June 2022; PMLR Volume 168, pages 234–247.

- [2] **L. Cothren**, A. Ospina, G. Bianchin, E. Dall’Anese. “Perception-based Online Optimization of Linear Time-Invariant Dynamical Systems.” *2022 56<sup>th</sup> Asilomar Conference on Signals, Systems, and Computers*, November 2022; pages 1357–1361.
- [1] **L. Cothren**, G. Bianchin, S. Dean, E. Dall’Anese. “Perception-based Sampled-data Optimization of Dynamical Systems.” *22<sup>nd</sup> Annual World Congress of the International Federation of Automatic Control*, July 2023; Volume 56, pages 5083–5088.

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## Conference Presentations

### Oral Presentations

- [P.4] **Cothren, L.K.**, Bianchin, G., Dean, S., and Dall’Anese, E.. Perception-based Sampled-data Optimization of Dynamical Systems. *22<sup>nd</sup> Annual World Congress of the International Federation of Automatic Control*, Yokohama, Japan, July 9–14, 2023.
- [P.3] **Cothren, L.K.**, Ospina, A., Bianchin, G., and Dall’Anese, E.. Perception-based Online Optimization of Linear Time-Invariant Dynamical Systems. *2022 Asilomar Conference on Signals, Systems, and Computers*, Palo Alto, CA, October 30–November 2, 2022.

### Posters

- [P.2] **Cothren, L.K.**, D’Souza, R., and Bradley, E.. Coupled catastrophes in systems with bidirectional feedback. *2026 Dynamics Days Conference*, Tucson, AZ, January 9–11, 2026. **\*Selected for best poster award.**
- [P.1] **Cothren, L.K.**, Bianchin, G., and Dall’Anese, E.. Data-enabled Gradient Flow as Feedback Controller: Regulation of Linear Dynamical Systems to Minimizers of Unknown Functions. *4<sup>th</sup> Annual Learning for Dynamics & Control Conference*, Palo Alto, CA, June 23–24, 2022.

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## Conference, Symposium and Workshop Participation

- 2026 **Attendee**, Santa Fe Institute: Complex Systems Summer School, Santa Fe Institute. June 2026
- 2026 **Microcredential**, Critical Dialogic Pedagogy: DEI-Informed Dialogic Pedagogy, Center for Teaching and Learning at University of Colorado Boulder. January–April 2026
- 2026 **Badge**, Research Communication and Presentation Badge, The Graduate School at University of Colorado Boulder. February 13, 2026
- 2025 **Attendee**, 2025 Dynamics Days, Denver, CO. January 3–5, 2025
- 2025 **Attendee**, 2025 SIAM Conference on Applications of Dynamical Systems, Denver, CO. May 11–15, 2025
- 2025 **Microcredential**, Foundations in High-Performance Computing, Office of Information Technology at University of Colorado Boulder. October 7, 2025
- 2023 **Badge**, Graduate Peer Mentoring Badge, The Graduate School at University of Colorado Boulder. August 2022–May 2023
- 2021 **Attendee**, 42nd International Summer School of Automatic Control of Grenoble, France, Grenoble, France. September 2021

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## Service and Outreach

- 2024–2025 **Tutor**, *High School Equivalency Test Preparation Classes*, Boulder Public Library, Boulder, CO.
- 2022–2024 **Graduate Peer Mentor**, *Graduate School Peer Mentoring Program*, University of Colorado Boulder, Boulder, CO.
- 2021–2023 **Fellow**, *Dean’s Future Leaders Fellowship*, University of Colorado Boulder, Boulder, CO.
- 2019–2021 **Member**, *Association for Women in Mathematics*, Arizona State University, Tempe, AZ.

- 2022 **Mentor**, *American Heart Association: STEM Goes Red Mentor*, University of Colorado Boulder, Boulder, CO.
- 2018–2022 **Panelist**, *Centennial High School STEM Panel*, Centennial High School, Peoria, AZ.
- 2018–2020 **Member**, *Fulton Ambassadors*, Arizona State University, Tempe, AZ.
- 2019–2020 **Member**, *Society for Women Engineers*, Arizona State University, Tempe, AZ.
- 2019 **Counselor**, *Engineering Summer Camp (E2)*, Arizona State University, Tempe, AZ.
- 2018 **Counselor**, *Engineering Summer Camp (E2)*, Arizona State University, Tempe, AZ.
- 2018 **Member**, *Engineering Projects in Community Service*, Arizona State University, Tempe, AZ.